

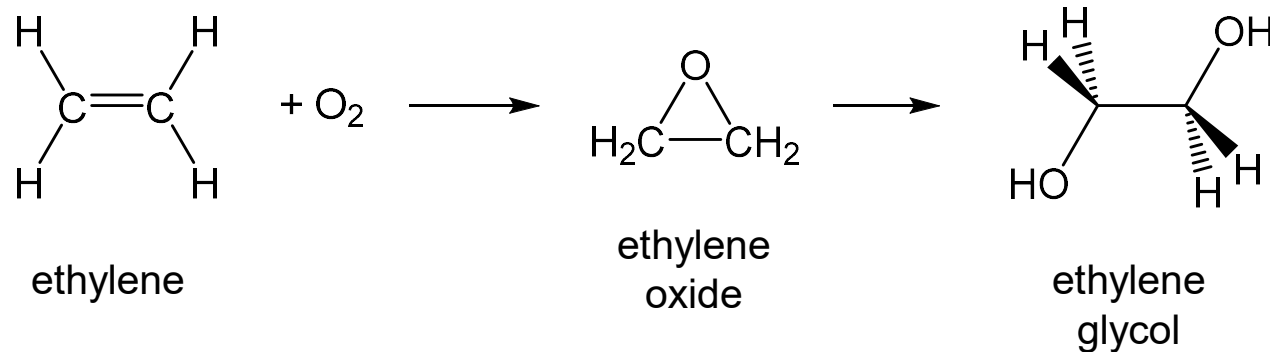
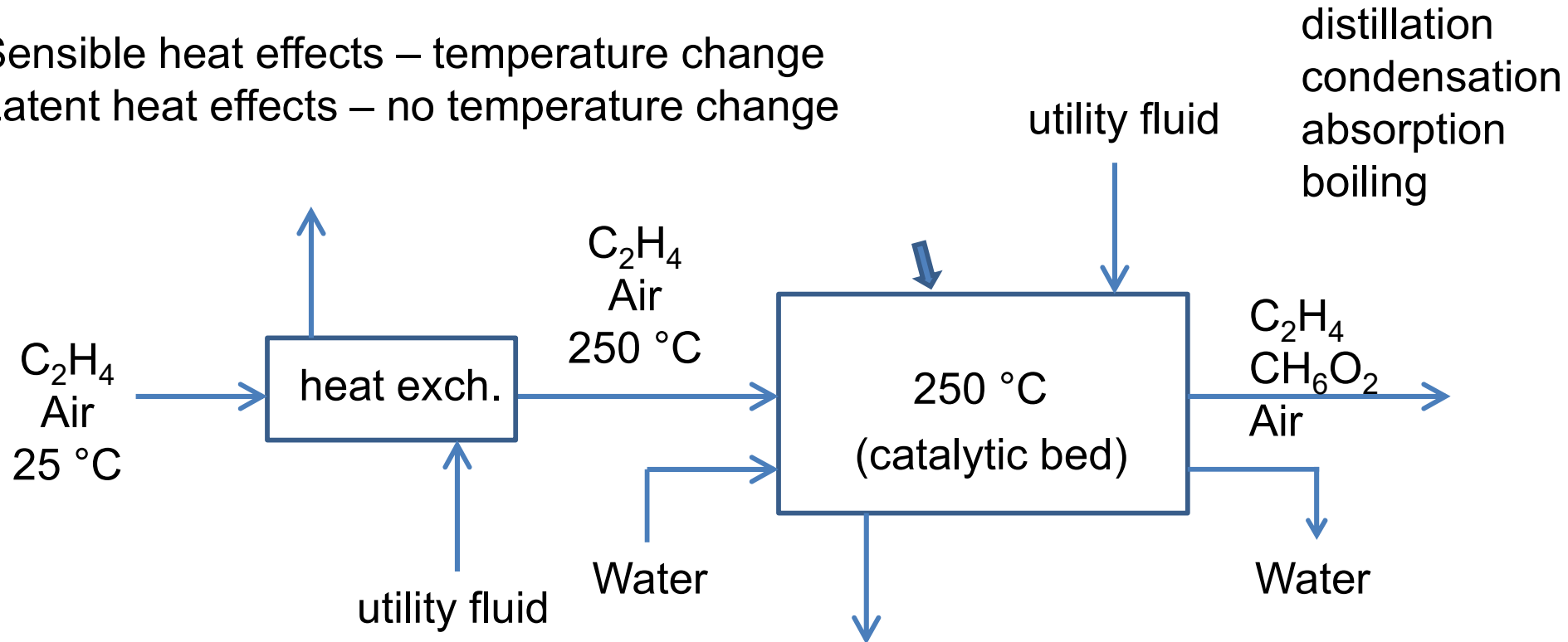
CH365 Chemical Engineering Thermodynamics

Lesson 17

Latent Heat and Heats of Reaction,
Formation, and Combustion

Ethylene Glycol Process

- Sensible heat effects – temperature change
 ➔ Latent heat effects – no temperature change



Latent Heats of Pure Substances

One component evaporating or condensing – No temperature change but heat is transferred. Latent heat of fusion / vaporization.

Latent Heats of Pure Substances

One component evaporating or condensing.

Latent Heats of Pure Substances

One component evaporating or condensing.

Standard Heat of Reaction

Heat effects in chemical reactions are manifestations of the differences in molecular structure.

There are a large number of chemical reactions, each of which can be carried out at an essentially infinite set of process conditions.

Tabulation of all possible heat effects for all possible reactions is impossible.

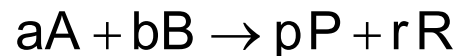
Heats are calculated from data for reactions conducted under standard conditions.

Heat associated with a chemical reaction depends on the temperatures of the reactants and products. A consistent standard treatment is that products and reactants at same temperature.

Standard Heat of Reaction

Tabulation of all possible heat effects for all possible reactions is impossible. Heats are calculated from data for reactions conducted under standard conditions.

Products and reactants are at the same temperature



Standard Heat of Reaction is the enthalpy change when a moles of A and b moles of B in their standard states at temperature T react to form p moles of P and r moles of R in their standard states at temperature T

Standard States:

Gases: ideal gas at 1 bar

Liquids and Solids: real pure liquid or solid at 1 bar



Degree symbol designates standard state

$C_p^{\circ} = C_p^{\text{ig}}$ for gases

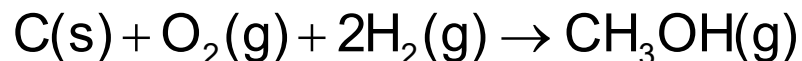
ΔH_{298}° Standard value at 298.15 K

Standard state properties are functions of T

Standard Heat of Formation

Tabulation of all possible heats of reaction is not practical. The standard heat of reaction can be calculated if the heats of formation of the reactants and products are known.

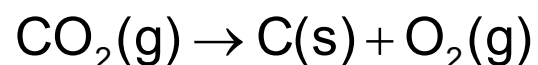
In a *formation reaction*, a compound forms from its constituent elements.



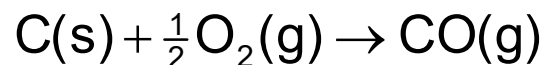
Heat of formation is based on one mole of product

Water-gas-shift reaction:

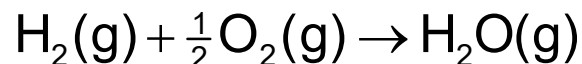
Table C.4, page 672



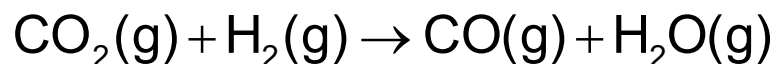
$$\Delta H_{298}^{\circ} = 393,509 \text{ J} \quad \text{negative in Table C.4 (listed as product)}$$



$$\Delta H_{298}^{\circ} = -110,525 \text{ J}$$



$$\Delta H_{298}^{\circ} = -241,818 \text{ J}$$



$$\Delta H_{298}^{\circ} = 41,166 \text{ J}$$

Questions?